SRS v0.4 Review — Issue List

Context-aware AAC Board · B - The Outliers

Item	Value
Reviewed document	Software Requirements Specification v0.4 (`docs/requirements/SRS.md`, commit `b0e5887`)
Review type	Internal sanity check before the peer-review comments
Reviewer	Jiahui Chen
Review date	2026-10-29
Checked against	Code and tests on `main` (108 tests passing), `docs/API.md`, git history, SRS v0.2
Result	24 issues: 7 high, 10 medium, 7 low. No changes made to the SRS

Summary

#	Location	Type	Severity	Issue
1	REQ-INT-004	Completeness	High	Reuse endpoint missing from the API requirement
2	3.5.8 Deadline	Completeness	High	No design (W12) or launch (W16) milestone
3	Figure 3, Appendix A	Correctness	High	Request shown as stored at confirmation time
4	REQ-FUNC-001/002/003, INT-001, PERF-001, AVAIL-001	Verifiability	High	Acceptance criteria not measurable
5	3.2 Functional	Completeness	High	"No suggestions" behaviour lost
6	Chapter 4, Table 56	Traceability	High	No requirement-to-test links
7	Chapter 4, 3.5.2, E.4, App. G, evidence fields	Consistency	High	Stale and generic evidence
8	1.2 Product Scope	Consistency	Medium	"Semantic caching deferred" vs implemented reuse
9	E.2, Figure 4, REQ-INT-004	Consistency	Medium	Endpoint count differs
10	3.6.2, 3.6.6, E.3	Traceability	Medium	Reuse and backup chain have no requirement ID
11	REQ-INT-001, INT-003, ML-003	Atomicity	Medium	Requirements bundle several conditions
12	REQ-INT-001, INT-003, FUNC-007, ML-005	Consistency	Medium	Confirmation rule stated four times
13	REQ-ML-003/004/005/006	Verifiability	Medium	Weak verification methods for AI
14	Owner fields	Consistency	Medium	Inconsistent or questionable owners
15	Chapter 3	Completeness	Medium	No accessibility requirement

#	Location	Type	Severity	Issue
16	Figures 2 and 3	Correctness	Medium	Sign-in drawn as a working step
17	REQ-SEC-005/007, 3.6.2	Completeness	Medium	Reuse history privacy not specified
18	Chapter 1, 1.1, 1.2, 2.1, 2.2	Clarity	Low	Duplicated sentences and a fragment
19	REQ-ML-004, E.4, 3.1.1	Clarity	Low	Typos and stray text
20	Table 5	Completeness	Low	Glossary missing used terms
21	Table 6 (R3, R4, R6)	Traceability	Low	References not locatable or outdated
22	Front matter, PDF	Form	Low	List of Tables and page numbers missing
23	Appendix B	Completeness	Low	Open decisions without owner or date
24	3.6.4, Appendix B	Completeness	Low	Interest-discovery proposal not referenced

High severity

Issue 1 — Reuse endpoint missing from REQ-INT-004

- **Location:** 3.1.3 / REQ-INT-004
- **Type:** Completeness
- **Finding:** The statement lists `POST /api/suggestions`, `POST /api/suggestions/image`, `PUT /api/requests/{request_id}/feedback`, `GET /health`, `GET /ready` and `/openapi.json`. `POST /api/suggestions/reuse` is merged on `main` and appears in Appendix E.3 and Figure 4, but not in the requirement. The acceptance criteria say "conform to docs/API.md", and `docs/API.md` is still v0.1 without the reuse endpoint.
- **Recommendation:** Add the reuse endpoint and the `error_code` field to the statement and acceptance criteria. Ask the backend owner to update `docs/API.md` to v0.2 first, so the criteria point at a correct contract.

Issue 2 — Deadline section has no design or launch milestone

- **Location:** 3.5.8 Deadline
- **Type:** Completeness
- **Finding:** Only "SRS submission in Week 7" is stated; all other dates are "TBD". The course milestones for the detailed design (Week 12) and the product launch and pitch (Week 16) are missing.
- **Recommendation:** Add a milestone table (W7 SRS, W12 detailed design, W16 launch and pitch) and a Mermaid `gantt` block, as the course GitHub guide recommends for schedules.

Issue 3 — Figure 3 stores the request at confirmation time

- **Location:** Figure 3 (AAC request activity diagram); Appendix A step 7
- **Type:** Correctness

- **Finding:** The step "Store request, suggestion and acceptance" follows "Explicitly confirm". In the code, the request and its suggestions are stored when they are generated (`app/services/suggestions.py`); confirmation only stores the feedback through `PUT /api/requests/{request_id}/feedback`.
- **Recommendation:** Add "Store request and suggestions" after generation and relabel the confirmation step "Store acceptance (PUT feedback)". Align Appendix A step 7.

Issue 4 — Acceptance criteria are not measurable

- **Location:** REQ-FUNC-001, REQ-FUNC-002, REQ-FUNC-003, REQ-INT-001, REQ-PERF-001, REQ-AVAIL-001
- **Type:** Verifiability
- **Finding:** FUNC-001 rejects "excessive inputs" without limits, although the code enforces 20 symbols, 80 characters per symbol and 500 characters per text. FUNC-002 says "oversize" while the code uses 5 MiB and 20 megapixels. FUNC-003 expects "1–3 options" while its statement allows 0. INT-001 says "all controls can be completed". PERF-001 and AVAIL-001 have no threshold (TBD).
- **Recommendation:** Copy the enforced limits and error codes into the acceptance criteria (SRS v0.2 had them). For PERF-001, propose a provisional p95 target from the measured 1.7–6.2 s Flash-Lite generations, marked "provisional until the AI evaluation".

Issue 5 — "No suggestions" behaviour has no requirement

- **Location:** 3.2 Functional
- **Type:** Completeness
- **Finding:** SRS v0.2 REQ-FUNC-004 required an empty list with a notice when no sentence fits. v0.3 reused the ID for "Stable request identity", so the behaviour is no longer specified, although the code implements it. For AAC users an honest empty answer is safer than a guessed sentence.
- **Recommendation:** Add REQ-FUNC-012 "Handle no suggestions", verified by `test_mock_does_not_misrepresent_free_text`.

Issue 6 — No requirement-to-test traceability

- **Location:** Chapter 4, Table 56
- **Type:** Traceability
- **Finding:** Every family row says "See individual entries", and no requirement names a test or artifact, although 108 automated tests exist and v0.2 linked each requirement to test names.
- **Recommendation:** Replace Table 56 with one row per requirement: method, test or artifact (real test names), current status and date.

Issue 7 — Stale and generic evidence

- **Location:** Chapter 4 (Table 55), 3.5.2, E.4, Appendix G, Evidence/status fields
- **Type:** Consistency
- **Finding:** The document still reports the Week 6 snapshot (26 tests; 19 passed, 8 in progress, 6 not run, 12 not started). Today 108 tests pass on Windows and the reuse endpoint is merged. Appendix G says the document "does not claim new test execution" while the AI evidence cites dated live measurements. 36 evidence fields end with "confirm against current test log and running frontend", including backend-only requirements (migrations, CI, secrets).
- **Recommendation:** Re-assess each status (Passed, Partial, Prototype, Not run, Not started) with a date. Replace the generic phrase with the concrete evidence. Update Appendix G.

Medium severity

Issue 8 — Scope contradicts the implementation

- **Location:** 1.2 Product Scope, Table 4
- **Type:** Consistency
- **Finding:** "Semantic caching" is listed as deferred, but reuse of accepted sentences is implemented (exact match in the backend, similarity in the AI prototype).
- **Recommendation:** Split the row: "Reuse of accepted sentences" in scope; "embedding-based semantic search" deferred.

Issue 9 — Endpoint count differs between sections

- **Location:** E.2, Figure 4, REQ-INT-004
- **Type:** Consistency
- **Finding:** E.2 says "six documented endpoints", Figure 4 says "7 endpoints", REQ-INT-004 lists six.
- **Recommendation:** Use 7 everywhere after fixing Issue 1.

Issue 10 — Reuse and backup chain have no requirement ID

- **Location:** 3.6.2, 3.6.6, E.3
- **Type:** Traceability
- **Finding:** Reuse of accepted sentences and the backup chain with quota handling are described only in prose, so Chapter 4 cannot trace them.
- **Recommendation:** Team decision: add REQ-ML-007 (backup models and quota handling) and REQ-ML-008 (reuse), or fold them into existing acceptance criteria. Adding IDs changes the count of 45.

Issue 11 — Requirements bundle several conditions

- **Location:** REQ-INT-001, REQ-INT-003, REQ-ML-003
- **Type:** Atomicity
- **Finding:** INT-003 combines photo input and audio output; INT-001 lists six actions; ML-003 covers intent, negation and prompt injection. One failing part fails the whole requirement, and the parts have different owners and methods.
- **Recommendation:** Split where owners or methods differ (for example INT-003a photo input, INT-003b speech output).

Issue 12 — Confirmation rule stated four times

- **Location:** REQ-INT-001, REQ-INT-003, REQ-FUNC-007, REQ-ML-005
- **Type:** Consistency
- **Finding:** "Speak only after confirmation" is written four times with different wording; a later edit could make them contradict each other.
- **Recommendation:** Keep the normative rule in FUNC-007 and reference it from the other three.

Issue 13 — Weak verification methods for AI requirements

- **Location:** REQ-ML-003, ML-004, ML-005, ML-006
- **Type:** Verifiability

- **Finding:** ML-003 (negation) and ML-004 (image flag) are verified by "Analysis / Demonstration", although offline tests exist for both. ML-005, a UI behaviour, lists "Analysis".
- **Recommendation:** Test for ML-003, ML-004 and ML-006; Demonstration for ML-005.

Issue 14 — Inconsistent or questionable owners

- **Location:** Owner fields
- **Type:** Consistency
- **Finding:** "Team", "All" and "Thibaut / team" are mixed. REQ-SEC-007 (consent and retention) is owned by "Abid / Veronika" although consent is a team decision. REQ-INT-005 is owned by Jiahui while the interface lives in the backend.
- **Recommendation:** One label for the whole team; SEC-007 owned by All; INT-005 owned by Jiahui / Veronika.

Issue 15 — No accessibility requirement

- **Location:** Chapter 3
- **Type:** Completeness
- **Finding:** The users have motor and communication impairments, but accessibility appears only as an open decision. No requirement covers target size, keyboard and touch use, contrast or screen readers.
- **Recommendation:** Add a usability/accessibility requirement (for example WCAG 2.1 AA for the three pages), marked "proposed" until the team agrees.

Issue 16 — Sign-in drawn as a working step

- **Location:** Figures 2 and 3
- **Type:** Correctness
- **Finding:** "Sign in" is the first step, but authentication is not implemented (REQ-SEC-005 planned); Appendix A says account verification is "not assumed".
- **Recommendation:** Label it "Sign in (interface only; authentication planned)" in both figures.

Issue 17 — Privacy of the reuse history not specified

- **Location:** REQ-SEC-005, REQ-SEC-007, 3.6.2
- **Type:** Completeness
- **Finding:** Accepted sentences are personal data and are reused across requests. The schema has no user ID, and no retention rule covers this history.
- **Recommendation:** Name the reuse history explicitly in SEC-005 (ownership) and SEC-007 (retention and deletion).

Low severity

Issue 18 — Duplicated sentences and a fragment

- **Location:** Chapter 1 introduction, 1.1, 1.2, 2.1, 2.2
- **Type:** Clarity
- **Finding:** The chapter 1 introduction describes the product twice; 1.2, 2.1 and 2.2 repeat their own opening sentences; 1.1 contains the fragment "Define the expected external behaviors…".
- **Recommendation:** Editing pass that keeps the meaning.

Issue 19 — Typos and stray text

- **Location:** REQ-ML-004, E.4, 3.1.1
- **Type:** Clarity
- **Finding:** The ML-004 title is lowercase ("non-deceptive operation"); an E.4 bullet contains a stray "(4)"; 3.1.1 still says "the supplied screens".
- **Recommendation:** Correct in the same editing pass.

Issue 20 — Glossary missing used terms

- **Location:** 1.3, Table 5
- **Type:** Completeness
- **Finding:** Scene hint, intention, feedback, generation chain, sentence provider (defined only in prose), HTTP 429/503 and TBD are used but not in the table.
- **Recommendation:** Add them to Table 5.

Issue 21 — References not locatable or outdated

- **Location:** 1.4, Table 6 (R3, R4, R6)
- **Type:** Traceability
- **Finding:** R3 ("supplied PDF") and R4 ("supplied PPTX") give no location. R6 points to `docs/VERIFICATION.md`, which still describes the v0.1 starter.
- **Recommendation:** Link the files in `weekly-reports/`; mark `VERIFICATION.md` as historical or update it.

Issue 22 — List of Tables and page numbers missing

- **Location:** Front matter; PDF export
- **Type:** Form
- **Finding:** v0.3 had a List of Tables; the Markdown edition dropped it. The PDF has no page numbers or running header.
- **Recommendation:** Restore the List of Tables; add page numbers to the PDF export.

Issue 23 — Open decisions without owner or date

- **Location:** Appendix B
- **Type:** Completeness
- **Finding:** Ten decisions are listed "before professor submission" without owner or due date, while submission is this week.
- **Recommendation:** Add an owner and target week to each; mark those that block the Week 12 design document.

Issue 24 — Interest-discovery proposal not referenced

- **Location:** 3.6.4, Appendix B
- **Type:** Completeness
- **Finding:** The proposal `docs/design/Interest_Discovery_Proposal.md` affects REQ-ML-003 but is not mentioned.
- **Recommendation:** One line in 3.6.4 and one open decision in Appendix B.

Suggested order of work

Step	Issues	When
1	1, 3, 5, 9, 16, 19	Before the peer-review comments arrive (small factual fixes)
2	2, 4, 6, 7, 8	Together with the peer-review comments
3	10, 11, 15, 17	After a team decision (they change requirements or their count)
4	12, 13, 14, 18, 20–24	Final pass before the next PDF export